Formal Architecture and Machinery of Appendices G, I, J, K, and L

An engineer-oriented guide for hardware architects and design-verification engineers

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G Contract	I Local replay	J Safety glue	K Liveness	L Witness selection

Based on: Catapult HLS User View Scheduling Rules, 18 July 2026

This guide is explanatory. The scheduling-rules document remains the normative source.

The latest version of the scheduling rules document is available here:

https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib_examples/doc/catapult_user_view_scheduling_rules.pdf

Audience

Readers comfortable with state machines, handshakes, FIFOs, assertions, and basic formal reasoning, but not necessarily specialists in trace semantics or mechanized theorem proving.

Contents

1. Executive orientation

PURPOSE	**Provide a single mental model for five appendices that otherwise appear as separate layers of formal detail.** **Engineer takeaway:** Treat the appendices as a proof stack: G specifies the contract, I generates local evidence, J constructs the global safety witness, K proves a separate conditional deadlock result, and L selects source witnesses when latency-sensitive choices matter.

The formal architecture is best understood as two related, but deliberately separate, proof pipelines.

- **Safety pipeline (G -> I -> J).** Shows that a covered reachable RTL behavior has a matching behavior in the prescribed source reference model at the chosen observable boundary. This is an observable trace-compatibility claim, not cycle-by-cycle or full-state equivalence.
- **Liveness/deadlock pipeline (G -> I -> J -> K).** Starts from the safety correspondence, reconstructs the wait-relevant cutpoint observation, and then rules out a specific class of reachable closed internal wait-cycle cutpoints under additional global progress and liveness premises.
- **Witness-selection adapter (L).** Supplies the WC witness needed by I and J when failed non-blocking polls or other latency-sensitive epsilon choices affect later observable behavior. It selects an admissible source execution aligned with the free-running RTL; it does not redefine the source model or stall the RTL.

KEY ARCHITECTURAL FACT: Safety and liveness are not one theorem

Appendix J establishes a global observable-safety correspondence from local evidence. Appendix K does not strengthen that correspondence into liveness automatically. It adds a different argument with different assumptions: wait-for-graph reconstruction, drain discipline, fairness/progress, cutpoint coverage, and the scheduler-robust source-liveness premise A5-L.

1.1 What the formal framework is trying to achieve

The user-facing goal is a verification flow in which most functional verification and debug can remain at the pre-HLS SystemC level, while the RTL is checked against a precise implementation contract. The formal appendices make that promise conditional and auditable: they identify exactly which source rules, implementation properties, witness choices, boundary abstractions, and liveness assumptions must hold.

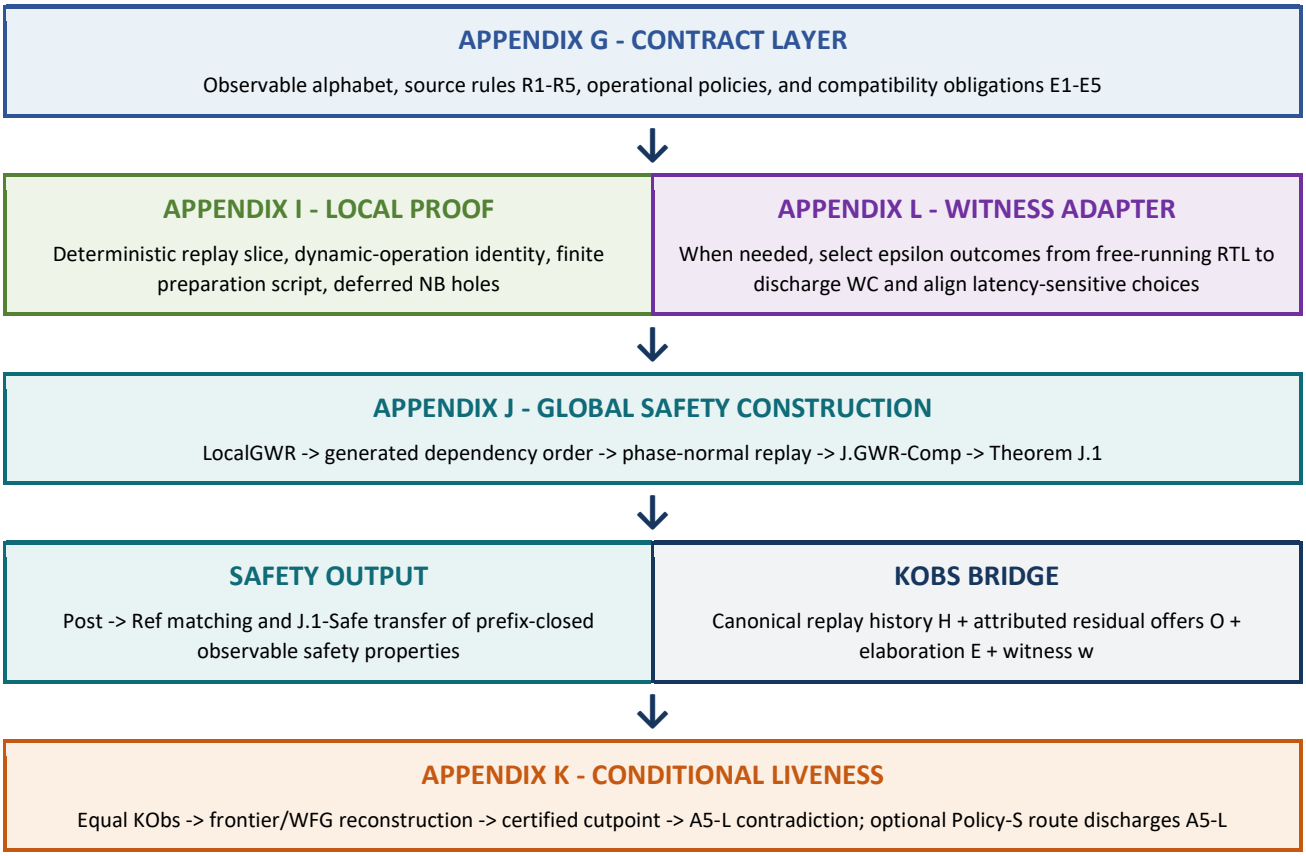
The central idea is to compare only architecturally meaningful observations: committed message transfers, synchronization operations, and anchored signal reads/writes at selected boundaries. Internal FSM states, pipeline movement, arbitration bookkeeping, and other micro-steps are represented as epsilon behavior unless they affect a named observable or proof predicate.

1.2 What the framework does not claim

- It does not claim identical cycle timing between pre-HLS and post-HLS models.
- It does not claim equality of all internal state, requests, or pipeline registers.
- It does not automatically prove that a particular HLS run or RTL instance satisfies E3, E4, B-faithfulness, R.RegSeq, or the other implementation obligations.
- It does not infer global deadlock freedom from the source-order no-reverse rule alone.
- It does not treat an arbitrary finite Policy-S simulation prefix as a complete liveness proof.
- It does not make latency-sensitive polling safe merely by hiding failed polls as epsilon steps; such sites require NB-CF, isolation, direct witness construction, or Appendix L snooping.

2. Architecture at a glance

PURPOSE	**Show the dependency chain from source/RTL semantics to the final safety and liveness conclusions.** **Engineer takeaway:** The load-bearing seam is between J and K: J exports one equal KObs record for matched cutpoints; K reasons only through that compact observation plus separate admissibility and liveness evidence.



2.1 The five interfaces between appendices

Interface	What crosses it
G -> I	G supplies the operational source model, observable labels, issue/commit rules, R.RegSeq, the Policy-L/Policy-S interpretation, and the E3 implementation discipline used by local replay.
I -> J	I supplies process-local replay/preparation facts packaged as J.PCert. These facts intentionally do not claim global channel enablement or one common source execution.
L -> I/J	L supplies a witness selector proving WC when epsilon-modeled choices can affect later observable control, values, request attribution, or frontier identity.
J -> K	J supplies a reachable source witness, phase-correct reachability, and one equal KObs record at matched normalized cutpoints. K does not compare arbitrary internal states.
Policy-S -> K	SDet, K.CompleteS, K.RespCov, and K.RespClean describe one selected deterministic conservative source run. K.2b/K.2c use it to discharge the global A5-L premise on the supported fragment.

2.2 Generic theorem machinery versus design-specific evidence

The framework deliberately separates proof algorithms that should be mechanized once from evidence that must be supplied or checked for each design, theorem instance, or RTL build.

Category	Examples
Reusable formal machinery	Definitions of trace compatibility, local-to-global construction, dependency ranking, phase normalization, KObs reconstruction, WFG extensionality, and the serial-dominance theorem.
Per-design source evidence	Source well-formedness, unique signal anchors, R.RegSeq classification, WC/NB-CF/snooping coverage, shared-memory lowering, barrier well-formedness, and value-determinism conditions.
Per-RTL/tool evidence	E3 scheduling conformance, E4 channel realization, exact capacities and boundaries, B-faithfulness, request attribution, reset behavior, and local process/channel/atomic certificates.
Per-liveness-instance evidence	The fixed capacities, elaboration, stimulus/environment, witness, simulator semantics, Policy-S run identity, response-monitor coverage/bounds, cutpoint coverage, and the selected A5-L discharge route.

3. Core vocabulary and proof boundaries

PURPOSE	**Define the small set of objects that repeatedly connect G, I, J, K, and L.** **Engineer takeaway:** Most confusion disappears once three distinctions are maintained: observable events versus epsilon micro-steps; committed history versus current residual offers; and safety correspondence versus liveness assumptions.

Term	Engineer-oriented meaning
Sigma (Σ)	The selected alphabet of observable events: committed Push/Pop transfers, synchronization events, and selected signal Read/Write events. The proof boundary determines which internal or DUT-boundary signals/channels are included.
Epsilon (ε)	Internal behavior not directly included in Σ: pipeline advances, internal FSM transitions, failed non-blocking polls, arbitration bookkeeping, and similar micro-steps, subject to WC and K-epsilon restrictions.
Observable unit	A singleton Σ event or an explicitly atomic same-cycle group such as a rendezvous Push/Pop, paired synchronization transaction, R2C fixed-point unit, signal-anchor group, or RESET unit.
Issue	The cycle attributed to a dynamic source message-operation instance when its endpoint-owned request begins. Issue attribution is operational evidence; it is not generally reconstructible from committed labels alone.
Commit	The clock cycle in which the selected boundary observer sees the operation complete; for message passing, both valid and ready are true and the payload is taken in that cycle.
Sys_ref	The prescribed source reference transition system: raw rendezvous Sys, capacity-annotated Sys_B, or elaborated Sys_B^elab with explicit staged/bundle/join/epoch-gate boundaries.
Horizon	The selected cycle/decision horizon used by J to group construction actions and enforce registered result availability and L1/L2/L3 phase ordering.
KObs	The compact cutpoint record <E,w,H,O>: elaboration E, witness w, canonical replay history H, and attributed residual offers O.
Residual offer	A typed current outstanding source-level request at an explicit abstract boundary, including process, frontier item, dynamic call, direction, and reset epoch.
WFG	Wait-for graph over internal processes. An edge P -> Q exists when a minimal pending request of P is disabled and Q owns the unique complementary endpoint.

CUTPOINT BOUNDARY: KObs is not full-state equivalence

KObs records enough information to reconstruct the Appendix K predicates: process replay slice, explicit channel state, pending/frontier classification, boundary requests, channel enablement, wait-for edges, and snapshot drain closure. It intentionally omits arbitrary internal RTL state and historical Issue timestamps. Therefore equal KObs is a carefully designed quotient, not a claim that Sys_ref and RTL_B are in identical states.

3.1 Safety direction: why Post -> Ref is the normal signoff argument

For ordinary safety signoff, the useful deductive direction is: every covered reachable RTL prefix has a matching reachable Sys_ref prefix with the same canonical observable history. Therefore, if all relevant Sys_ref histories satisfy a prefix-closed safety property, the RTL histories do as well. This is Corollary J.1-Safe.

The reverse direction is intentionally restricted. A particular source prefix is guaranteed to have an RTL realization only when it is already equipped with an RTL-consistency annotation package, or when Appendix L selects the source witness to align with the actual RTL choices. The document does not claim unrestricted equality of the source and RTL behavior sets.

4. Appendix G - the observable scheduling contract

PURPOSE	**Define what behaviors are observable, which source schedules are legal, and what the post-HLS implementation must preserve.** **Engineer takeaway:** Appendix G is the semantic contract. It does not by itself construct a matching source execution; Appendices I and J do that work.

4.1 Source-side rules R1-R5

Rule	Role in the architecture
R1 - synchronization order	Synchronization calls of each process remain in dynamic source order and commit in strictly increasing cycles.
R2 - signal anchoring	For a sequential process, a signal Read is logically anchored to the closest preceding synchronization; a signal Write is anchored to the closest succeeding synchronization.
R2C - combinational fixed point	Combinational signal observations are emitted at a deterministic unique epsilon-fixed point for the explicitly modeled combinational component.
R3 - sync isolation	A synchronization boundary separates the message operations in the immediately preceding and succeeding source intervals.
R4 - safe issue order	Dynamic message operations preserve source issue order, with a prefix-closed reading: a later operation cannot issue unless every required earlier operation has issued.
R5 - channel capacity semantics	The source reference uses rendezvous or finite-capacity channel semantics at the selected explicit abstract boundaries; elaboration may introduce proof-internal channels but does not reorder ordinary source calls.

DEFINITION: R.RegSeq - registered sequential-interface semantics

For a sequential process, request and Push-data outputs in cycle t are determined from registered process/pipeline state and operands available before the cycle-t boundary. A result returned at cycle t cannot create a dependent request in that same cycle; the first permitted issue is a later cycle.

- **Why it matters.** It eliminates same-Horizon decision-to-dependent-request edges that would be inconsistent with clocked hardware.
- **Where it is used.** Appendix I local preparation, J.FoundationRank/J.DepRank, J.RegPhaseNF, and the Appendix K serial-reduction assumptions.
- **DV implication.** The flow needs evidence that RTL request/data outputs are registered as assumed and that zero-time dependent non-blocking cascades are rejected, isolated, or modeled separately.

4.2 Operational issue policies

Appendix G treats the source model as an operational transition system with explicit local process state, explicit channel state, dynamic operation identities, pending requests, and epsilon micro-state. Two source issue policies are important:

Policy	Meaning and use

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Policy L - liberal	The most permissive source issue policy used to construct the safety witness and to define the scheduler-robust liveness premise A5-L. It may issue reached operations subject to source order, R.RegSeq, channel semantics, and the L0-L4 phase rules.
R.LAdm - admissible Policy L	Adds the liveness-specific obligations: request nonwithdrawal, K-epsilon, and the R.16/K.Drain discipline once a settled fully disabled minimal frontier is exposed.
Policy S - serial blocking	A conservative process-facing policy in which a later blocking operation in the same source interval does not issue until required earlier blocking operations have committed in a strictly earlier cycle. It is used as a deadlock-stress execution in Appendix K.

DEFINITION: R.LAdm - L0/L1/L2/L3/L4 phase discipline

Each cycle has a process-owned pre-settle window (L1), one epsilon-settle snapshot with requests frozen (L2), boundary decisions and commits (L3), and registered result availability for the next cycle (L4). If L2 exposes a fully disabled minimal frontier, no new source-later work may be launched and only finite drain-only progress is allowed.

4.3 Post-HLS compatibility obligations E1-E5

Obligation	What the RTL/tool flow must establish
E1	Preserve each process's synchronization-event order.
E2	Preserve anchored signal values/visibility for sequential processes and fixed-point signal units for combinational processes.
E3	Preserve safe message Issue order and attribution, including same-interface order, distinct-interface no-reverse order, and the prefix-closed Issue condition.
E4	Realize legal rendezvous or finite-capacity FIFO behavior at each selected explicit abstract channel boundary, including payload/order/occupancy behavior.
E5	Do not move a message operation across its immediate synchronization boundary: predecessor operations issue no later than the sync; successor operations issue strictly later.

BOUNDARY CONDITION: B-faithfulness and Sys_B^elab

E4 only has meaning at a chosen abstract boundary. The flow must show that every transfer crosses that boundary, occupancy and enablement match the declared capacity B(c), internal movement is epsilon-opaque, and no hidden grant, throttle, backpressure, join, or epoch-gate condition changes availability. If ordinary Sys_B cannot express the realization, Sys_B^elab must expose the additional boundaries explicitly.

4.4 Trace compatibility is observational, not temporal equivalence

The relation approximately-equal (≈) matches dynamic occurrences and values at the selected alphabet while preserving the explicit constraints above. Independent observations may occur in different cycles or be grouped differently in the two executions. The relation is directed and annotation-bearing; it is not a symmetric mathematical equivalence over bare label sequences.

Appendix G also separates the explanatory causal-dependency graph from the well-founded order used by Appendix J. Causal dependency captures information flow and may contain legal same-cycle atomic cycles. Appendix J therefore constructs a narrower required-order relation and a separate generated dependency graph for the proof induction.

READING RULE: What to take from Appendix G

For architects: G defines what implementation freedom remains invisible to a latency-insensitive environment. For DV: G identifies the observable checker boundary and the conformance obligations. For formal implementers: G defines the transition-system vocabulary that I, J, and K consume.

5. Appendix I - per-process replay and preparation

PURPOSE	**Show that each process can locally replay the selected RTL observable prefix and prepare for its next participation using a finite source-owned script.** **Engineer takeaway:** Appendix I is intentionally local. It identifies the right dynamic operation and reaches its request/decision point, but it does not assert that the complementary endpoint is present or that all process scripts can be merged globally.

5.1 Inputs and preconditions

Input	Why I needs it
Fixed initial state and stimulus	The process is replayed under the same global Sigma-causal environment behavior, not an arbitrary P-local stimulus.
WC witness	Every epsilon-modeled choice that can affect later observable control, values, identities, requests, or frontiers is fixed consistently.
I.A0 / E3	The RTL message-issue discipline is supplied as an implementation assumption, including prefix-closed issue attribution.
R.RegSeq	Dependent next-cycle requests cannot be created from same-cycle returned results.
B4 / finite pipeline depth	Internal micro-progress has a finite epsilon bound when the process is not externally stalled.
B2/B3 when progress is invoked	Continuously enabled actions and complementary channel discharge are eventually selected; these are environment/scheduler assumptions, not consequences of source code.
Source well-formedness	Signal anchors, dynamic operation identities, direct-input contracts, shared-memory lowering, and reset behavior are well formed.

5.2 The replay slice: I.Obs, I.Obs-Congruence, I.DR, and I.DR-prime

The key simplification is to replay only the source-state facts that can affect later observable behavior or Appendix K's request/frontier predicates. I.Obs includes control location, future-relevant variables, dynamic operation identities, pending/frontier/request attribution, relevant witness state, and anchored values. It ignores diagnostic state and epsilon-only timing state that cannot influence the proof interface.

LEMMA: I.Obs-Congruence

If two epsilon-quiescent source states have the same Sigma-relevant slice and take the same next observable label under the same witness, their successor Sigma-relevant slices remain equal. This is the one-step closure property that makes deterministic replay possible.

LEMMA: I.DR and I.DR-prime - deterministic replay

I.DR establishes equality of the Sigma-relevant source slice after the same observable prefix at normalized cutpoints. I.DR-prime gives the corresponding immediate post-unit state, which is important because Appendix J constructs the next Horizon without automatically normalizing after every unit.

5.3 Deferred non-blocking decisions

A process-local proof cannot decide a non-blocking poll from process state alone because success or failure depends on the common incident-channel snapshot and possible complementary requests. Stopping the local replay at every poll would be too restrictive: a registered process may execute independent same-Horizon work that does not depend on the poll result.

DEFINITION: I.DeferredNBHole

A deferred hole records the dynamic non-blocking call, explicit boundary, snapshot key, Horizon, result destinations, reset epoch, witness-selected expected outcome, and NoDependentUse evidence. The process-local preparation script can carry the unresolved hole through independent work while forbidding any dependent use of its result.

5.4 The local preparation theorem

LEMMA: I.3 - finite process-local preparation script

Given the replay state, exact dynamic operation identity, R.RegSeq, WC, E3/I.A0, source well-formedness, and typed footprints, I.3 constructs a finite process-owned epsilon/request path to the selected operation, its persistent issued request, or a carried deferred non-blocking decision hole. The script does not commit an unmatched observable unit and does not assume global enablement.

- **Output.** The information later packaged in J.PCert: replay state, dynamic identity, finite per-Horizon actions, request/pending attribution, stable payload/value facts, footprints, and deferred-hole records.
- **Non-result.** No complementary endpoint, common channel snapshot, globally legal merge, or reachable system extension is proved here.
- **Why finite.** The source-control segment is finite and internal progress is bounded by B4/FPD when not externally stalled.

5.5 Engineer example: independent work after a poll

Suppose a process performs a PopNB, updates an independent diagnostic counter, prepares a request on another endpoint whose control and payload do not depend on the PopNB result, and only later branches on the poll status. Appendix I may carry the poll as a DeferredNBHole through the independent counter/request work. Appendix J later freezes the complete channel request snapshot at L2, decides the poll once globally, and prevents the branch from occurring until the result is available under R.RegSeq.

ARCHITECTURAL SEAM: Why Appendix I cannot be the whole safety proof

Local scripts may each be valid when viewed against read-only summaries, yet still conflict through shared channel state, atomic rendezvous pairing, a common non-blocking snapshot, reset scope, or overlapping field accesses. Appendix J proves that the scripts coexist and derives a legal global order; it does not merely conjoin them.

6. Appendix J - local-to-global safety construction

PURPOSE	**Construct one reachable Sys_ref execution from local process/channel/atomic certificates and prove observable compatibility with a selected RTL prefix.** **Engineer takeaway:** Appendix J is the main safety theorem. Its defining achievement is that global witness realizability is derived from local evidence rather than accepted as an unexplained system-level premise.

6.1 Theorem J.1: the top-level safety result

THEOREM: J.1 - execution-level / trace-set compositional compatibility

For every covered reachable RTL_B prefix, the normal compositional route uses Definition J.LocalGWR and Theorem J.GWR-Comp to construct a reachable Sys_ref prefix and a global witness package whose annotated projections satisfy E1-E5. A directly validated J.GWR package is an alternate route. The reverse Ref -> Post clause applies only to source prefixes already equipped with an RTL-consistent annotation/witness package.

Three details are important for engineering use:

- **Finite reachability, not only fair executions.** The Post -> Ref construction applies to finite reachable RTL prefixes; fair extension is a separate premise when an infinite execution is needed.
- **Required-prefix induction.** The proof orders observable units by the constraints that must be preserved, rather than by raw RTL cycle buckets or full source order across independent endpoints.
- **Annotated carrier.** Compatibility includes dynamic identities, witness choices, Issue/Commit attribution, request facts, and channel histories. It is stronger than matching an unannotated list of labels, but weaker than full-state equivalence.

6.2 The local certificate package

Component	What it certifies
J.PCert	Per-process replay and finite preparation evidence produced from Appendix I.
J.CCert	Per-explicit-channel boundary identity, capacity, E4 history, occupancy/head, payload, request snapshot, enablement, reset, and B-faithfulness evidence.
J.ACert	Agreement for explicitly atomic units: rendezvous pairs, paired synchronization, E2/R2C groups, RESET, and other named multi-participant actions.
J.EnvCert / J.ResetCert	Environment and reset-unit identity, scope, action, and locality evidence.
Typed footprints	Field-sensitive ReadSet/WriteSet evidence for every construction action, including component-local elaboration actions.
LocalAgree	Equality, uniqueness, ownership, coverage, and duplicate-field agreement across local records.
Generated dependencies	Locally justified Dep_J edges and enough evidence to prove that omitted pairs commute; the package does not supply a convenient global rank.

DEFINITION: J.LocalGWR

The complete local package for one RTL prefix: PCert for every process, CCert for every explicit boundary, ACert for every atomic unit, applicable environment/reset records, complete action/deferred-hole inventory, field-sensitive footprints, R.RegSeq and source conformance, generated dependencies, LocalAgree, and coverage. It must not assume that these records already form a realizable global replay.

6.3 How J derives a global construction order

J converts the certificate package into a finite graph of primitive construction actions. The graph includes process epsilon/control actions, request/attempt assertions, settled-snapshot decisions, complete commits, and reset/environment actions. Atomic groups are contracted before ordering.

Definition / lemma	Role
J.DepJ	Generated direct dependency edges from process control/data/Issue/ReturnedAtCycle rules, request-before-snapshot rules, channel/FIFO state, atomicity, reset/environment order, and explicit component dependencies.
J.DepSound	Every generated edge is a real operational prerequisite.
J.DepComplete	If two non-atomic actions are left unordered, they either commute by field-sensitive framing or are the explicitly proved same-cycle buffered Push/Pop dual update.
J.FoundationRank	A graph-independent lexicographic semantic rank based on reset epoch, Horizon, semantic stage, local ordinal, and component tie data.
J.DepRank	Every generated edge strictly advances the foundation rank.
J.DepAcyclic	A directed cycle would strictly increase the well-founded rank back to its start, which is impossible.
J.CanonicalRank	A derived deterministic totalization of the acyclic prerequisite relation. Stable tie keys order only commuting actions.

LOAD-BEARING RESULT: J.DepComplete + J.DepRank

The flow cannot simply provide a global rank that is assumed correct. It must generate every non-commuting prerequisite from local rules, prove that unordered actions really commute, and prove each edge advances an independent semantic foundation rank. This is what makes the construction auditable and prevents a hidden global replay assumption.

6.4 Registered phase normalization

Within each Horizon, the proof must respect actual clocked interface semantics. It cannot decide a non-blocking result and then create a dependent request in the same cycle, and it cannot evaluate different polls against inconsistent partial channel snapshots.

LEMMA: J.RegPhaseNF - registered phase normal form

Every finite same-Horizon construction fragment can be permuted, without changing its canonical observable history, into: L1 process evaluation and request/attempt writes; one L2 settled common snapshot; and L3 non-blocking decisions plus complete boundary commits. Registered results become available only in L4/next cycle.

LEMMA: J.NBDecision

Once all L1 predecessors have executed and J.CCert supplies the common settled snapshot, each deferred non-blocking call has exactly one legal source result. A failed poll cannot coexist with a matching enabled rendezvous request, and a successful poll cannot be invented in a full/empty state that forbids it.

6.5 From one Horizon to the complete source witness

Lemma / theorem	Function in the construction
J.Frame	A certified transition changes only its WriteSet; disjoint state and guards are preserved.
J.LocalLift	Embeds one Appendix I process-local script into a reachable global state when owned/shared locations match its certified summaries.
J.PrepareCommute	Moves independent preparation actions into the common L1 window while respecting snapshots and true dependencies.
J.UnitEnable	After L1/L2, the requests, payloads, channel state, atomic participants, and holes make each selected L3 unit genuinely enabled.
J.UnitExt	Executes one complete Horizon bucket, updates replay/channel/atomic state, resolves holes, and preserves the construction invariant.
J.GWR-Comp	Outer induction over Horizon buckets constructs the complete reachable Sys_ref witness chain and the stable J.GWR interface.

THEOREM: J.GWR-Comp - local certificates construct global witness realizability

Starting from J.LocalGWR, the theorem generates and validates the dependency graph, derives the canonical order, normalizes each Horizon to the registered L1/L2/L3 form, executes the buckets through J.UnitExt, and produces both Definition J.GWR and J.RegPhaseReach for the same reachable source witness.

- **Non-circularity.** LocalGWR supplies no global order, no phase-normal replay, and no global extension premise.
- **Stable interface.** J.GWR can also be supplied directly by a translation validator or symbolic replay tool, but that direct route must independently validate the same reachability and ordering interface.
- **Reset handling.** RESET is an atomic observable unit; RW1-RW5 and Lemma J.RS splice a new matched reset epoch while framing unaffected components.

6.6 Proposition J.0 is deliberately separate

PROPOSITION: J.0 - pairwise composition lemma

Given an already selected compatible source/RTL trace pair, compatible per-process annotated projections, and per-channel E4 well-formedness, J.0 lifts E1-E5 to the system level. It does not construct the source trace, prove B-faithfulness, or establish J.GWR. Keeping it separate prevents circular reasoning.

6.7 Safety transfer for the verification flow

COROLLARY: J.1-Safe - source-side safety transfer through Post -> Ref

For a coverage domain of RTL prefixes equipped with valid J.GWR evidence, any prefix-closed safety predicate on canonical DUT-boundary histories transfers from all reachable Sys_ref prefixes to every covered RTL prefix. One complete pre-HLS run is sufficient only when J.RefProjCover proves that its canonical observable prefixes cover all relevant reachable source observations.

6.8 The J-to-K cutpoint bridge

Trace matching alone is insufficient for deadlock reasoning. Two executions can have the same committed history while one endpoint has not reached a blocking call and the other has an attributed persistent request. J therefore augments the canonical replay history with the terminal residual-offer set.

Bridge element	Purpose
J.HCanon	Proves that the selected source and RTL prefixes have the same canonical replay history H modulo the permitted epsilon/history quotient.
J.RegPhaseReach	Records that the exact source witness was constructed with R.RegSeq and the Horizon L1/L2/L3 phase discipline.
J.OfferReach	Shows that <E,w,H,O> is actually realized by a reachable normalized Sys_ref cutpoint, not merely a well-typed tuple.
J.OfferConf	Bidirectionally and uniquely matches every residual offer in O to a relevant asserted RTL request and every relevant RTL request back to exactly one offer.
J.KObsConf	Binds one exact source/RTL/cutpoint package and establishes equality of the typed KObs record.
Corollary J.1	Exports equal KObs and therefore equal reconstructed frontier, request, channel, enablement, WFG, and snapshot drain predicates to Appendix K.

AUDIT CRITICAL: J.OfferConf request-to-offer completeness

Every relevant asserted RTL boundary request must be attributed to exactly one source residual offer at the selected explicit boundary. Omitting a hidden asserted request, duplicating attribution, or using an unrelated source cutpoint invalidates the KObs bridge. This is the implementation abstraction boundary on which Appendix K relies.

7. Appendix K - conditional deadlock preservation

PURPOSE	**Preserve absence of reachable drain-closed closed internal message-passing wait cycles from the selected source-reference instance to covered RTL cutpoints.** **Engineer takeaway:** K is not a generic progress theorem. It proves a specific reachable-cutpoint property and is conditional on A5-L plus fairness, drain, environment, value, lowering, and coverage obligations.

7.1 The deadlock predicate

Appendix K evaluates epsilon-quiescent cutpoints at explicit point-to-point channel boundaries. The WFG contains only internal message-passing dependencies; environment-facing channels and SyncChannel barriers are excluded from the graph and handled through separate scope/assumption rules.

Object	Meaning
Front_P	The minimal pending blocking message-operation items of process P whose endpoint requests are currently asserted. Merely being the next source call is not enough; the request must have issued.
Blocked process	Front_P is non-empty and every frontier item is ChannelDisabled at the selected explicit boundary.
WFG edge P -> Q	A disabled frontier request of P waits on the unique complementary endpoint owner Q.
DrainClosedNow(D)	No already-offered non-frontier request owned by D remains enabled; otherwise finite downstream drain could still change the candidate wait structure.
Dead_K / Dead_K^W	A non-empty closed strongly connected blocked component with no outgoing internal WFG edge and snapshot drain closure, excluding declared observationally closed waivers.

SCOPE: Reachable closed internal wait-cycle cutpoint

The operative predicate is stronger than “the system is currently slow” and different from a global permanent-deadlock definition. It identifies a reachable internally closed wait cycle at a drain-closed snapshot. Escape through a nondefault timed/reactive environment co-frontier is a separate concern controlled by K.EnvMF and explicit waivers.

7.2 Reconstructing the wait state from J.KObs

Result	Function
Lemma K.0	Reconstructs the operational minimal pending blocking frontier from KObs. It distinguishes a reached-but-unissued candidate from an asserted persistent request.
Lemma K.1a	Characterizes disabled Push/Pop frontiers for buffered and rendezvous channels at the chosen explicit boundaries.
Lemma K.2	WFG extensionality: equal KObs implies exactly equal reconstructed wait-for graphs.
Lemma K.2a	Snapshot drain-closure extensionality: equal KObs implies equal DrainClosedNow for any process set.
R.21 reconstruction lemmas	Factor process replay, channel state, offers, frontier, enablement, WFG, drain closure, and the deadlock predicate into pure functions of KObs.

7.3 One certified cutpoint versus universal coverage

DEFINITION: K.CertifiedCutpoint

A K-ready package for one finite reachable RTL prefix and one selected normalized epsilon-quiescent cutpoint. It binds the theorem instance and contains the J.1/KObs correspondence, the exact source witness, J.RegPhaseReach, K.DrainReach, point-to-point boundary facts, waiver closure, and the other K-facing assumptions for that same package.

THEOREM: K.1A - certified-cutpoint preservation from A5-L

Assume one certified RTL cutpoint satisfies the prohibited deadlock predicate. Equal KObs transfers the same frontier/WFG/drain predicate to the exact reachable Sys_ref witness. J.RegPhaseReach plus K.DrainReach prove that witness is R.LAdm-admissible. This contradicts A5-L. Therefore that certified RTL cutpoint cannot contain the prohibited internal wait cycle.

COROLLARY: K.1A-Total - universal result under K.CertCov

K.1A is pointwise. To claim that no reachable K-relevant RTL cutpoint has the prohibited wait structure, K.CertCov must establish that every reachable prefix/selected normalized cutpoint has a complete K.CertifiedCutpoint package.

7.4 The central source-liveness premise A5-L

PREMISE: A5-L - scheduler-robust liberal source liveness

No finite Policy-L prefix admissible under R.LAdm for the selected Sys_ref instance reaches an epsilon-quiescent non-waived Dead_K cutpoint. Because Policy-S-shaped serial prefixes are included among admissible Policy-L prefixes, a design with a reachable serial wait cycle makes A5-L false; another proof technique cannot certify the unchanged instance.

A5-L is intentionally global. Internal wait cycles depend on interactions among processes, capacities, environment behavior, and the selected issue policy. Unlike the J safety construction, it cannot generally be reduced to independent per-process liveness theorems.

7.5 The primary Policy-S discharge route

The practical route avoids exploring all liberal schedules directly. It uses one fully specified deterministic conservative source simulator and proves that any liberal internal deadlock would cause a finite, covered response failure on that same selected serial execution.

Element	Role
K.Low / Sys_K	Normalize synchronization waits and witness-selected non-NB-CF dependencies into explicit blocking guard/epoch-gate channels so the serial argument operates on literal blocking frontiers.
SDet	Fix every history-affecting choice: model, capacities, elaboration, reset, stimulus/environment, witness, arbitration, concrete simulator scheduling, Policy-S issue semantics, and epsilon normalization. Select one run artifact rho_S^det.

Element	Role
K.EnvMF	For multi-frontier components, require StandardEnv/B1-available plus the default ready output environment; timed/reactive or otherwise nondefault environments are restricted to a certified single internal frontier.
K.RespCov	Default-total coverage of every static internal process-facing blocking site and relevant dynamic class, with a finite bound or justified disposition.
K.RespClean	No finite prefix of the complete selected run witnesses a non-waived timeout, maximal-finite pending frontier, cancellation/reset failure, or end-of-test pending failure.
K.CompleteS	The selected run is maximal and fair. A terminating run reaches a declared terminal state; a nonterminating run requires invariant/model-checking/lasso or equivalent completeness evidence.
A5-S / CF-S	The package K.CompleteS + K.RespCov + K.RespClean for the SDet-selected run, bound to one certificate header/trust boundary.

LEMMA: K.2b - deterministic serial-source dominance with response-failure extraction

Under the blocking/lowering, value-determinism, environment, fairness, drain, and coverage premises, any R.LAdm-admissible Policy-L internal deadlock forces the same SDet-selected Policy-S execution to produce a finite non-waived K.RespFail witness. The theorem does not claim the two runs have the same trace or the same deadlock component.

COROLLARY: K.2c - the selected deterministic Policy-S execution suffices

If the complete selected Policy-S run has total response coverage and is response-clean, K.2b rules out any liberal deadlock counterexample; therefore A5-S implies A5-L on the supported fragment. There is no quantification over alternate Policy-S schedulers and no confluence theorem is required.

PRACTICAL WARNING: A finite clean prefix is not enough

For a terminating design, the run must reach a declared maximal terminal state with all covered monitors resolved. For a nonterminating design, a finite simulation trace is only evidence; a complete invariant, sound model-checking result, recurrence/lasso proof, or equivalent argument must establish that response failures never occur.

7.6 Alternate direct discharges of A5-L

Route	When it can discharge A5-L
K.Str-1 - acyclic dependency graph	If the full static process dependency graph is acyclic, no WFG cycle can occur.
K.Str-2 - ranked dependency	A well-founded rank strictly decreases along every realizable WFG edge, excluding cycles.
K.Str-3 - credit/token cycle breaking	Every static dependency cycle contains an edge proved unrealizable at reachable cutpoints, often by occupancy, initial-token, or credit invariants.
CF-3 / direct invariant	An inductive abstraction proves the prohibited Policy-L cutpoint unreachable.
CF-4 / exploration	Sound exhaustive or bounded model checking over the selected transition system.
CF-5 / tool certificate	A tool-emitted certificate using knowledge of the scheduled control graph, storage, arbitration, joins, and automatic flush.

These are alternate proofs of exactly the same A5-L property. They cannot rescue a design for which an admissible serial-shaped Policy-L prefix already reaches Dead_K.

7.7 Standing assumptions A1-A11 in engineer terms

Assumption	Plain-language role
A1	The source and RTL satisfy the previously defined R-rules and B-rules at the selected boundaries.
A2	Finite internal epsilon depth / finite stutter when not externally stalled.
A3	Weak fairness: a continuously enabled action is eventually selected.
A4	Channel progress: persistent complementary requests eventually discharge full/empty/rendezvous blocking.
A5	For the user-facing Policy-S route, a complete bounded-response-clean selected serial run; K.1A itself consumes A5-L directly.
A6	Point-to-point channels with unique producer and consumer at every evaluated boundary.
A7	B1 RTL determinism/witness selection plus the exact equal-KObs source/RTL correspondence and R.LAdm qualification.
A8	Barrier participation is well formed; barrier mismatches are outside the internal message-passing deadlock theorem.
A9	At most one Push and one Pop commit per channel per cycle; multi-lane/burst resources must be modeled explicitly.
A10	Every buffered channel is logically empty after each reset boundary.
A11	K-epsilon: hidden epsilon steps do not create, remove, or redirect the WFG-relevant blocking structure.

THEOREM: K.1 - user-facing deterministic Policy-S preservation

Combine uniform K.1A premises, K.CertCov, SDet, K.CV, applicable NB-Align/K.Low, point-to-point boundaries, nonwithdrawal, K-epsilon, K.Drain, K.EnvMF, K.RespCov, and A5-S. Corollary K.2c supplies A5-L; K.1A-Total then excludes the prohibited reachable internal wait-cycle cutpoint in RTL_B.

7.8 What Appendix K does not cover

- Barrier mismatches or conditional SyncChannel participation (handled by A8 and separate protocol verification).
- Waiting indefinitely for an external environment input, output readiness, termination, EOF, cancellation, or reset behavior not represented by the internal WFG predicate.
- Arbitrary starvation or throughput guarantees unrelated to the defined blocking frontier and response-monitor schema.
- Cycle-accurate RTL response bounds; K.Resp evidence is a source-side discharge of A5-L, not an RTL timing contract.
- Designs whose correctness requires favorable eager/same-cycle issue when a serial-shaped admissible prefix deadlocks.

8. Appendix L - witness selection and snooping

PURPOSE	**Align latency-sensitive epsilon choices in a source simulation with the unique free-running RTL behavior so that WC can be discharged.** **Engineer takeaway:** Snooping selects one admissible source witness. It must not feed pre-HLS readiness back into RTL, relabel payloads, drop/reorder observations, or pretend that uncovered latency-sensitive choices are irrelevant.

8.1 Why snooping is needed

A non-blocking arbiter, interrupt controller, or retry/timeout block may make a later observable choice based on which request arrived first or how many polls failed. HLS changes internal latency, so the raw pre-HLS and post-HLS simulations can select different epsilon outcomes even though both satisfy the broad scheduling contract. When those outcomes affect later Sigma-visible behavior, Appendix I needs a specific WC-compatible witness.

If NB-CF holds - failed non-blocking outcomes do not affect later observable control, values, frontier, or request attribution - the witness is trivial and snooping is unnecessary. Cadence-sensitive polling is different: it is a localized latency-sensitive protocol and must be isolated or explicitly snoop-aligned.

DEFINITION: L.1 - witness selector / snooping

At each relevant epsilon-modeled source choice, the selector returns the outcome chosen by the matched RTL prefix using only causally available history. It cannot depend on future RTL events or invent a retry/timeout outcome inconsistent with the isolated protocol.

LEMMA: L.2 - snooping establishes WC

If the snooped set covers every epsilon choice that can affect later observable behavior, the wrapper changes only epsilon-choice selection, selected outcomes are causal and RTL-consistent, successful and failed non-blocking calls retain the required Sigma/epsilon interpretation, and cadence-sensitive blocks are isolated, then the selected source execution satisfies WC for Appendices I and J.

8.2 One-directional architecture

CONDITION: L.Dir - one-directional alignment / RTL-side non-interference

The RTL_B execution is free-running under the fixed stimulus. The pre-HLS snoop sequence must consume the recorded RTL snoop sequence in ordinal order, with matching kind and payload. No pre-HLS backpressure or readiness may stall or alter RTL. An L.Dir violation means the compared run is outside the discharged relational scope and requires triage.

8.3 Wrapper realization requirements W1-W5

Requirement	Engineering meaning
W1 - recorded ordinal observation	Capture each RTL snooped event by dynamic ordinal and hold it until the pre-HLS side consumes the same ordinal. Do not rely on overlapping live assertion windows.

Requirement	Engineering meaning
W2 - gate only commit enable	Gate only the mirror signal that enables the pre-HLS commit: observed valid for a pre-HLS Pop or observed ready for a pre-HLS Push. Never modify the source endpoint-owned request.
W3 - registered sampling	Register the RTL observation before it enters the source-side gate to avoid delta-cycle races; the extra source latency is part of witness selection.
W4 - compare and watchdog	At each source snoop commit, compare ordinal, kind, and payload against the recorded RTL event. Never substitute the RTL payload into the source, because that would hide value divergence. Detect missing, extra, reordered, or never-consumed observations.
W5 - independent coverage proof	The wrapper mechanism is sound only for the points it records. A separate analysis must prove that all epsilon choices capable of affecting later observables are covered.

IMPLEMENTATION PITFALL: Why a live AND is unsound

ANDing the current pre-HLS and current RTL handshake levels can drop an RTL event when the RTL assertion retires before the source arrives. It can also reorder or duplicate matching under multiple outstanding events. The normative rule is an AND against a recorded per-ordinal observation retained until source consumption.

8.4 Harness soundness and shared testbenches

LEMMA: L.5 - online realization of the witness selector

A co-simulation harness realizes the abstract selector only when it preserves the free-running RTL trace, implements W1-W5, satisfies L.Dir, and produces the same source-side witness choices assumed by the formal comparison. The formal theorem is about the fixed RTL trace and selected source witness; harness correctness is an additional implementation obligation.

A shared reactive testbench becomes problematic once one model can lag significantly, because reactions may be driven by whichever model the testbench observes. The flow should replicate the stimulus/testbench per model, or structure a shared environment around a common causal transaction stream rather than raw cycle timing.

8.5 What snooping means formally

Snooping does not use RTL to change the source specification. The source model is intentionally under-specified with respect to permitted latency and epsilon choice. The RTL observation selects one admissible execution of that specification for comparison. This is analogous to supplying an environment parameter or arbitration witness after observing the implementation, provided the selected behavior was already allowed by the source semantics.

8.6 Latency-robustness stress testing

Appendix L also recommends randomized stalls, varied channel latency/capacity, and adversarial weakly fair scheduling in the pre-HLS model. This is not a substitute for the formal premises, but it is a practical way to detect designs that accidentally rely on incidental ordering of causally independent events. A failure under such stress usually indicates that the source specification itself is outside the intended latency-insensitive model or that an explicit snoop/isolation boundary is missing.

9. How the appendices work together in a verification flow

PURPOSE	**Translate the theorem architecture into a concrete signoff workflow.** **Engineer takeaway:** Safety-only use stops after the J.1/J.1-Safe package. Appendix K is added only when the claim includes the defined internal wait-cycle freedom, and it requires substantially more global evidence.

9.1 Step-by-step safety flow

Step	Deliverable
1. Fix the theorem instance	Select Sys_ref/Sys_B/Sys_B^elab, explicit boundaries, capacities, observable alphabet, reset interpretation, fixed stimulus/environment, witness provenance, and tool/RTL identities.
2. Establish G-level conformance	Check source R-rules, signal anchors, R.RegSeq, E3, E4, E5, elaboration, and B-faithfulness.
3. Discharge WC	Use NB-CF, Appendix L snooping/L.2, direct witness construction, or a complete mixed per-site coverage argument.
4. Produce local evidence	Generate J.PCert from Appendix I and per-channel/atomic/environment/reset certificates plus typed footprints and local dependencies.
5. Run J.GWR-Comp	Derive the dependency order, registered phase-normal construction, reachable source witness, and J.GWR interface.
6. Apply Theorem J.1	Obtain Post -> Ref observable compatibility for the covered reachable RTL prefix.
7. Transfer the safety property	Use J.TraceCertCov and Corollary J.1-Safe over the required RTL coverage domain; use J.RefProjCover only when claiming one selected source run covers all source observations.

9.2 Additional liveness flow

Step	Deliverable
8. Choose normalized cutpoints	Use B4/B5 and record the exact source/RTL normalization witnesses.
9. Build the KObs bridge	Discharge J.HCanon, J.OfferReach, J.OfferConf, and J.KObsConf for the same source/RTL/witness/cutpoint package.
10. Qualify the source witness	Use theorem-derived J.RegPhaseReach plus explicit K.DrainReach to prove K.RLAdmReach.
11. Certify the cutpoint	Assemble K.CertifiedCutpoint including point-to-point boundaries, waiver closure, and all K-facing assumptions.
12. Discharge A5-L	Use the primary SDet + A5-S/CF-S + K.2c route, or a valid structural/invariant/exploration/tool route.
13. Establish coverage	Apply K.1A to one cutpoint; add K.CertCov for K.1A-Total or the universal user-facing K.1 claim.

9.3 Recommended reading order in the scheduling-rules document

- Read Appendix G first for the contract and operational policies.
- Read the Common Formal Definitions next, especially KObs, B-rules, and the side-condition discharge record.
- Read Appendix I to understand exactly what is local and why deferred non-blocking holes are needed.

- Read Appendix J as the central safety construction; focus first on J.LocalGWR, J.DepRank/J.DepAcyclic, J.RegPhaseNF, J.UnitExt, J.GWR-Comp, Theorem J.1, and Corollary J.1.
- Read Appendix L before K when witness-selected non-blocking or latency-sensitive interfaces are present.
- Read Appendix K for the deadlock predicate, KObs reconstruction, certified-cutpoint theorem, A5-L, and the Policy-S discharge route. Appendix K-P is the deeper proof companion for serial reduction.

10. Assumptions and evidence matrix

PURPOSE	**Summarize the named assumptions that most often determine whether a theorem instance is usable.** **Engineer takeaway:** The important audit question is not merely "is the assumption plausible?" but "what artifact establishes it for this exact source, RTL, capacity/elaboration, stimulus, witness, reset, and environment instance?"

Premise / evidence	Where it matters and typical discharge
B1 - RTL deterministic trace	Consumed by J/K/L selected-trace reasoning. Evidence: RTL/tool determinism proof, qualification, deterministic arbitration/environment setup, or a validated witness-selection setup.
B2 - weak fairness	Consumed by I progress embedding and K drain/progress arguments. Evidence: scheduler/arbiter fairness proof, assertions, or Appendix N pattern qualification.
B3 - channel progress	Consumed by I.2 and K. Evidence: persistent complementary endpoint requests plus E4/B2 or a per-boundary progress certificate.
B4/B5 - finite epsilon closure	Consumed by I local finite preparation and J/K normalized cutpoints. Evidence: finite pipeline/FSM depth and a unique quiescent combinational fixed point.
WC	Consumed by I and J; provenance may be NB-CF, L.2 snooping, direct witness construction, or mixed per-site coverage.
R.RegSeq	Consumed by I/J/K. Evidence: sequential interface structure, registered request/data outputs, and no same-cycle dependent request from a returned result.
B-faithfulness	Consumed by J/K at every explicit abstract boundary. Evidence: back-annotation, structural RTL checks, interface monitors, refinement checks, or tool certificates.
J.LocalGWR	Consumed by the normal J.GWR-Comp route. Evidence: complete local process/channel/atomic/env/reset records, footprints, dependencies, agreement, and coverage.
J.OfferReach/Conf/KObsConf	Consumed by Corollary J.1 and all K cutpoint reasoning. Evidence must bind the exact same source/RTL prefix, witness, history, offers, and normalized cutpoints.
K.Drain and K-epsilon	Consumed by K.RLAdmReach, K.1A, and serial dominance. Evidence: automatic-flush or equivalent drain-only structural proof plus no hidden WFG-changing epsilon behavior.
A5-L	Consumed directly by K.1A. Evidence: Policy-S route through K.2c or a valid structural/direct/invariant/exploration/tool proof.
SDet + A5-S	Consumed by K.2b/K.2c/K.1. Evidence: selected functional simulator identity, complete run proof, total response-monitor coverage, finite bounds, and response cleanliness.
K.CV / NB-Align / K.Low	Consumed by the serial route when values, non-blocking choices, sync waits, or lowering affect future operations. Evidence: explicit causal/value determinism and local lowering correspondence.
K.EnvMF	Consumed by the serial route. Evidence: StandardEnv/B1-available for multi-frontier components, or certified single-internal-frontier structure for nondefault environments.

Premise / evidence	Where it matters and typical discharge
L.Dir and W1-W5	Consumed by the online snooping harness. Evidence: free-running RTL, ordinal recording/consumption, comparison/watchdog, noninterference, and complete choice-point coverage.

10.1 A practical certificate boundary

A useful implementation is a versioned package with a common header that binds every artifact to one theorem instance: source and RTL build IDs, selected Sys_ref/Sys_K, boundaries/capacities/elaboration, observation projection, stimulus/environment/reset model, witness provenance, simulator/scheduler/arbitration identity, and checker/schema versions. Separate payloads then carry J safety certificates, CF-0 cutpoint correspondence, and one declared A5-L discharge route.

SIGNOFF RULE: Do not mix witnesses or cutpoints

A J.OfferReach proof for one source state, a J.OfferConf proof for another RTL cycle, and a K.Drain proof for a different witness do not compose. The formal architecture repeatedly requires exact identity of the theorem instance, stimulus, elaboration, witness, H, O, source prefix, RTL prefix, and normalized cutpoints.

11. Common misunderstandings

Misreading	Correct interpretation
“Trace compatible” means cycle accurate.	No. It preserves selected labels, values, ordering, atomicity, channel semantics, and anchors. Independent events may move between cycles.
The source pass transfers by Ref -> Post.	For ordinary safety, the deductive path is Post -> Ref plus J.1-Safe: RTL adds no covered canonical safety behavior absent from Sys_ref. Ref -> Post is restricted to annotated RTL-consistent or witness-selected source prefixes.
Per-process replay certificates already imply a system replay.	No. Shared channel state, snapshots, atomic participants, resets, and field interference can conflict. J.GWR-Comp proves the merge.
The causal-dependency graph is the J induction order.	No. Causal dependency explains information flow and may contain legal atomic cycles. J uses generated construction dependencies plus a foundation-rank proof.
KObs is a normalized execution or full state.	No. It is a typed observation quotient <E,w,H,O> sufficient for K-facing reconstruction.
No-reverse scheduling proves deadlock freedom.	No. It prevents one deadlock-introduction mechanism. Timing overlap, finite capacity, pipeline drain, environment behavior, and serial-shaped source deadlocks require Appendix K.
A clean finite Policy-S simulation proves A5-S.	Only if it is a declared maximal terminal run with all monitors resolved. A nonterminating design needs complete invariant/model-checking/lasso or equivalent evidence.
Another A5-L route can rescue a serial deadlock.	No. If an admissible Policy-S-shaped Policy-L prefix reaches Dead_K, A5-L is false for the unchanged instance.
Snooping changes the specification to match RTL.	No. It selects one behavior already admitted by the under-specified source latency/epsilon semantics.
A live AND is enough for snooping.	No. The wrapper must record per-ordinal RTL events and retain them until source consumption; otherwise events can be dropped or reordered.

Misreading	Correct interpretation
K.Resp is an RTL latency guarantee.	No. It is source-side evidence used to discharge A5-L through serial dominance.
Tool generation automatically discharges the proof obligations.	No. The tool or flow must emit documentation, certificates, static checks, monitors, or formal evidence for each named obligation.

12. Quick-reference index

PURPOSE	**Provide a compact map from theorem/definition names to their role in the proof architecture.** **Engineer takeaway:** Use this section as a navigation aid when reading the normative scheduling-rules document.

12.1 Appendix G

Name	Role
R1-R5	Source scheduling and channel-capacity contract.
R.RegSeq	Registered next-cycle dependence rule.
R.LAdm	Policy-L legality/admissibility and L0-L4 cycle phases.
Policy S	Conservative serial-blocking issue policy.
E1-E5	Post-HLS trace-compatibility obligations.
Causal dependency	Design-intent/information-flow relation, not the J induction order.

12.2 Appendix I

Name	Role
I.A0	RTL E3 message-issue discipline assumption.
I.Obs	Sigma/K-relevant source-state slice.
I.Obs-Congruence	One-step closure of the replay slice.
I.DR / I.DR-prime	Deterministic source replay after a matched prefix.
I.DeferredNBHole	Carried unresolved shared non-blocking decision with NoDependentUse.
I.3	Finite process-local preparation script.
I.4	Construction of the J.PCert records.

12.3 Appendix J

Name	Role
J.LocalGWR	Complete local certificate package; no global replay assumption.

Name	Role
J.DepJ / DepComplete / DepRank / DepAcyclic	Generate, complete, rank, and prove well-founded the construction dependencies.
J.RegPhaseNF	Normalize same-Horizon actions to registered L1/L2/L3 phases.
J.NBDecision	Decide each deferred non-blocking call from the common settled snapshot.
J.UnitExt	Extend one complete Horizon bucket.
J.GWR-Comp	Construct J.GWR and J.RegPhaseReach from local certificates.
J.0	Compose an already selected compatible pair.
J.1	Top-level observable compatibility theorem.
J.1-Safe	Transfer prefix-closed safety through Post -> Ref.
J.HCanon / OfferReach / OfferConf / KObsConf	Build the exact equal-KObs cutpoint bridge.
Corollary J.1	Export equal KObs and reconstructed K-facing facts.

12.4 Appendix K

Name	Role
Dead_K / Dead_K^W	Reachable drain-closed closed internal wait-cycle predicate.
K.0	Frontier reconstruction from KObs.
K.2 / K.2a	WFG and snapshot drain-closure extensionality.
K.RLAdmReach	Exact source witness is Policy-L admissible: RegPhaseReach + DrainReach.
K.CertifiedCutpoint	Complete package for one K-relevant RTL cutpoint.
K.CertCov	Package coverage for every reachable selected cutpoint.
A5-L	No admissible Policy-L prefix reaches Dead_K.
SDet	One fully specified deterministic Policy-S simulator/run.
K.RespCov / RespClean / CompleteS	Total bounded-response evidence over the complete selected run.
K.2b / K.2c	Serial dominance and A5-S -> A5-L reduction.
K.1A / K.1A-Total / K.1	One-cutpoint, universal, and user-facing liveness-preservation results.

12.5 Appendix L

Name	Role
L.1	Causal witness selector for epsilon-modeled source choices.
L.2	Coverage/noninterference conditions under which snooping proves WC.
L.3	NB-CF identity witness; no snooping required for irrelevant failed polls.
L.4	Cadence-sensitive polling requires isolation/snoop alignment.
L.Dir	Ordinal-preserving source consumption of a free-running RTL snoop stream.

Name	Role
L.5	Online harness realization of the abstract witness selector.
W1-W5	Recorded ordinal observation, commit gating, registered sampling, comparison/watchdog, and separate coverage proof.

12.6 One-sentence architecture summary

SUMMARY: The proof stack in one sentence

Appendix G defines the observable and operational contract; Appendix I proves finite process-local replay/preparation under a fixed witness; Appendix J derives a globally reachable source witness and trace-compatible safety result from local certificates; Appendix L supplies that witness when latency-sensitive epsilon choices require implementation-aligned selection; and Appendix K uses J's equal-KObs cutpoint plus separate admissibility and source-liveness evidence to exclude the specified internal wait-cycle structure in covered RTL cutpoints.

Source note: theorem and definition names follow the 18 July 2026 scheduling-rules document. This companion intentionally simplifies prose but does not replace the normative statements, side conditions, or proof text.